Database Management

Section Review

Database Migrations with Alembic

- A **database migration** is a file that describes a change to the database schema.
- Alembic is a standard migration tool for the SQLAlchemy ORM.
- Run **uv run alembic init alembic** to initialize Alembic. The final argument is the folder that will store the migrations.

The Migration Process I

- Update one or more models with changes (new fields, deleted fields, constraints, etc).
- Run **uv run alembic revision --autogenerate -m** with a revision message.
- Alembic will generate the migration in the **versions** directory. Each migration will have a unique ID.

The Migration Process II

- Double-check the migration for accuracy. Alembic makes mistakes.
- Update the **script.py.mako** file to revise the template for future migrations.
- A migration file is a *planned* change. The database has not been updated yet.

Upgrading

- Run **uv run alembic upgrade head** to run all pending migrations.
- Provide a number (ex., +1) after **upgrade** to move a certain number of migrations forward.
- Alternatively, provide the unique migration ID.

Downgrading

- **alembic downgrade -1** rolls back the previous migration. Customize the value to move multiple migrations backwards.
- **alembic downgrade id** accepts the ID of the migration to roll back to.
- **alembic downgrade base** rolls back all migrations, returning to the beginning of the schema.

Transactions

- A **database transaction** is a “unit of work” that groups multiple operations.
- The transaction can be committed (**commit**) or rolled back (**rollback**).
- The **session** object from SQLAlchemy creates a transaction behind the scenes. It rolls back automatically in case of error.

Database Indexes

- A **database index** is a data structure that accelerates lookup/join operations on a column's values by keeping track of their occurrences behind the scenes.
- In excess, database indexes can slow down write, update, and delete operations. Use them for columns that are frequently queried like foreign keys.
- A uniqueness constraint will create an index automatically.